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5. Three-Chain Equivalence (C2)

We adopt the *sign-compensated* ("σ_ϵ") scheme throughout C2: all rank-one bulk reductions are written in terms of $\mathrm{sgn}(\Sigma_\epsilon)\, I_T \equiv \sigma_\epsilon I_T$ so that patches with $\Sigma_\epsilon < 0$ are handled uniformly, $\mathrm{sgn}(\Sigma_\epsilon)\, I_T = -6\eta^2|\Sigma_\epsilon|$.

We prove that three ostensibly different quadratic routes— (i) a *rank-one determinant route* built out of the canonical traceless matrix $\mathcal{T}^\mu{}_\nu$, (ii) a closed-metric rank-one deformation, and (iii) the PT-even shadow extracted from the CS/Nieh–Yan chain— collapse, *after* C1 and projection, to the *same* bulk invariant $A_\star\sqrt{-g}\,\mathrm{sgn}(\Sigma_\epsilon)\, I_T$ up to a total derivative. Differences are carried entirely by improvement currents $\nabla_\mu J^\mu$ (closed forms on FRW/weak-field are in Appendix C).[3]

5.1. Preliminaries: Canonical Rank-One Objects And Normalization

Recall $\Sigma_\epsilon \equiv \Pi_{\mathrm{PT}}[(\partial\epsilon)^2]$, $\hat{n}_\mu \equiv \dfrac{\partial_\mu\epsilon}{\sqrt{|\Sigma_\epsilon|}}$, and the dimensionless, traceless rank-one matrix

$$\mathcal{T}^\mu{}_\nu \equiv \mathrm{sgn}(\Sigma_\epsilon)\Big(\hat{n}^\mu\hat{n}_\nu - \tfrac{1}{4}\delta^\mu{}_\nu\Big), \qquad \mathrm{Tr}\mathcal{T} = 0, \quad \mathrm{Tr}(\mathcal{T}^2) = \tfrac{3}{4}, \tag{22}$$

together with the dimension-one spurion scale (after C1),

$$\tau \equiv \hat{n}^\mu T_\mu \;=\; 3\,\eta\;\mathrm{sgn}(\Sigma_\epsilon)\,\sqrt{|\Sigma_\epsilon|}. \tag{23}$$

Key relation.

$$\tau^2 = 9\eta^2|\Sigma_\epsilon| = \frac{3}{2}\,(-\,\mathrm{sgn}(\Sigma_\epsilon) I_T), \qquad \mathrm{sgn}(\Sigma_\epsilon)\, I_T \equiv \mathrm{sgn}(\Sigma_\epsilon) I_T = -6\eta^2|\Sigma_\epsilon|. \tag{24}$$

5.2. Three One-Line Propositions (NY Split $\to *$ & Projection $\to$ Coefficient Match)

Proposition 1 (NY split). *$T^A \wedge T_A = \mathrm{NY} + e^A \wedge e^B \wedge R_{AB}$ with $\mathrm{NY} \equiv d(e^A \wedge T_A)$.*

Proposition 2 (PT-even shadow after $*$ and projection). *Under A2, the combined PT preserves orientation and satisfies $[\mathcal{PT}, *] = 0$. Applying $*$ to Prop. 1 and projecting to scalars, the term $*\mathrm{NY}$ contributes only as a boundary convention (A5), while the remaining PT-even bulk piece reduces, at quadratic order and* after C1, *to*

$$\delta^2\mathcal{L}_{\mathrm{CS}^+} = A_\star\,\sqrt{-g}\;\mathrm{sgn}(\Sigma_\epsilon)\, I_T + \nabla_\mu J^\mu_{\mathrm{CS}^+} \quad \textit{for some } A_\star \in \mathbb{R}.$$

Proposition 3 (Coefficient match under the σ_ϵ scheme). *With $\mathrm{Tr}\mathcal{T} = 0$, $\mathrm{Tr}(\mathcal{T}^2) = \frac{3}{4}$, and $\tau = 3\eta\;\mathrm{sgn}(\Sigma_\epsilon)\sqrt{|\Sigma_\epsilon|}$, both (i) the rank-one determinant route $\sqrt{\det[\mathbf{1} + (\frac{2}{3})\lambda\,\tau\,\mathcal{T}]}$ and (ii) the closed-metric deformation $\tilde{g}_{\mu\nu} = g_{\mu\nu} + (\frac{2}{3})\lambda\,\tau\,\hat{n}_\mu\hat{n}_\nu$ yield the same quadratic bulk coefficient $\delta^2\mathcal{L} = A_\star\,\sqrt{-g}\;\mathrm{sgn}(\Sigma_\epsilon)\, I_T$ with $A_\star = \lambda^2/8$.*

Proof (paper-level). Expand $\sqrt{\det(\mathbb{1} + X)} = 1 + \frac{1}{2}\mathrm{Tr}X + \frac{1}{8}[(\mathrm{Tr}X)^2 - 2\mathrm{Tr}X^2] + \mathcal{O}(X^3)$; with $X = (\frac{2}{3})\lambda\tau\mathcal{T}$, $\mathrm{Tr}\mathcal{T} = 0$ gives $\delta^2\sqrt{\det(\mathbb{1} + X)} = -\frac{1}{4}\mathrm{Tr}(X^2) = -\frac{1}{4}(\frac{4}{9})\lambda^2\tau^2\mathrm{Tr}(\mathcal{T}^2) = -\lambda^2\tau^2/12$. Using (24), $\delta^2\mathcal{L} = \sqrt{-g}(\lambda^2/8)\;\mathrm{sgn}(\Sigma_\epsilon)\, I_T$. The closed-metric route has the same Jacobian $\sqrt{-\tilde{g}} = \sqrt{-g}\sqrt{\det[\mathbb{1} + (\frac{2}{3})\lambda\tau\mathcal{T}]}$, hence the identical coefficient.

[3] We use "rank-one determinant route" (formerly "DBI"-type) to denote the volume deformation $\sqrt{\det[\mathbf{1} + (\frac{2}{3})\lambda\,\tau\,\mathcal{T}]}$ in our posture; this is *not* Born–Infeld gravity.

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